Interpreting IMF2 on a log(1+z) Axis (HHT)

Concepts, mappings to redshift space, and energy projections

1) Why use log(1+z)?

Redshift z relates to the scale factor a by $a = 1/(1+z)$.

Taking $x = \ln(1+z)$ turns multiplicative changes in $(1+z)$ (or equivalently $1/a$) into additive displacements.

Instantaneous frequency measured in x has a natural interpretation as cycles per unit $\ln(1+z)$, i.e., a multiplicative scale in a.

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x = ln(1+z)   ↔   a = e^{-x}
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On the x-axis, equal intervals Δx correspond to equal factors in $(1+z)$.

Thus, an oscillation with frequency $f(x)$ (cycles per unit x) captures features occurring at nearly constant multiplicative scale in cosmic expansion.

2) IMF / HHT essentials on x = ln(1+z)

EMD separates a signal $s(x)$ into intrinsic mode functions (IMFs).

For an IMF_k(x), the Hilbert transform produces the analytic signal A_k(x) e^{iφ_k(x)} with instantaneous amplitude A_k and phase φ_k.

The instantaneous frequency is ω_k(x)=dφ_k/dx and f_k=ω_k/2π.

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IMF_k(x)  →  z_k(x) = A_k(x) e^{i φ_k(x)}
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ω_k(x) = dφ_k/dx
f_k(x) = ω_k(x) / (2π)    [cycles per ln(1+z)]
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<u>Note</u>: Bandwidth is the range of instantaneous frequencies an IMF occupies.

One cycle corresponds to Δx = 1/f, mapping to a multiplicative change in (1+z) by e^{Δx}.

3) Mapping frequency and bandwidth to redshift space

If an IMF has local frequency f at redshift z ($x = \ln(1+z)$), one oscillation spans $\Delta x \approx 1/f$, which maps to:

$$\Delta x = 1/f \quad \Rightarrow \quad \Delta z \approx (1+z)(e^{1/f} - 1)$$

Higher f means finer structure in redshift (smaller Δz);
lower f means broader modulations (larger Δz).

4) Energy projection and Hilbert spectrum on x

For IMF_k, the Hilbert spectrum $H_k(x,f)$ assigns energy $|A_k(x)|^2$ at instantaneous frequency $f_k(x)$.

Summing over k gives $H(x,f)$. Integrate over f to get energy vs x, or over x for the marginal spectrum vs f.

$$H_k(x, f) \approx |A_k(x)|^2 \cdot \delta(f - f_k(x))$$

$$\begin{aligned} H(x, f) &= \Sigma_k H_k(x, f) \\ E(x) &= \int H(x, f)\, df \\ M(f) &= \int H(x, f)\, dx \end{aligned}$$

5) IMF2 interpretation in log(1+z) space

- IMF2 often captures the first non-trivial oscillatory mode after the trend; lower frequency than IMF1 but still localized.

- IMF2's $f_2(x)$ maps to $\Delta z \approx (1+z)(e^{1/f_2}-1)$.
  Coherent f_2 with high amplitude $A_2(x)$ indicates a repeatable modulation at a characteristic multiplicative scale in $(1+z)$.

- In your HHT screens, an IMF2 novelty around $z \approx 0.38$–0.70 would be consistent with BAO residual leverage in that range (seen via LOOCV).

6) Mapping back to cosmic evolution

Since $x = \ln(1+z) = -\ln a$, uniform frequency in x corresponds to uniform frequency in ln a.

An IMF with nearly constant f describes processes repeating per e-fold in the scale factor. For time interpretation, one may map z→t via a background H(z).

$$t(z) = \int_z^\infty dz' / [(1+z') H(z')]$$
$$H(z) = H0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

7) What can be learned from IMF2

- Localization of structured residuals in specific redshift bands.
- Scale diagnosis via f-range → Δz-range (fine vs broad structure).
- Cross-dataset checks: compare IMF2 energy in BAO vs SN residuals.
- Model stress tests where IMF2 energy and χ^2 leverage coincide (e.g., $z \approx 0.38$–0.70).

Appendix: Key equations and definitions

$$x \equiv \ln(1+z)$$
$$a = e^{-x}$$

Analytic signal:

$$z_k(x) = IMF_k(x) + i \cdot H[IMF_k(x)] = A_k(x)\, e^{i \varphi_k(x)}$$

Instantaneous frequency:

$$f_k(x) = (1/2\pi)\, d\varphi_k/dx \quad [\text{cycles per } \ln(1+z)]$$

Redshift span per cycle:

$$\Delta z \approx (1+z)(e^{1/f_k}-1)$$

Hilbert spectrum:

$$H(x,f) = \Sigma_k\, |A_k(x)|^2\, \delta(f-f_k(x))$$

$$\text{Energy vs } x: \quad E(x) = \int H(x,f)\, df$$

$$\text{Marginal vs } f: \quad M(f) = \int H(x,f)\, dx$$